Course Code: BCE-C523
Course Name: DESIGN AND ANALYSIS OF ALGORITHM

MM: 100 **Time: 3 Hr.** **L T P** **3 1 0**	**Sessional:30** **ESE:70** **Credit :4**

Prerequisites:	Basic knowledge of data structure
Objectives:	• Analyze the asymptotic performance of algorithms. • Write rigorous correctness proofs for algorithms. • Demonstrate a familiarity with major algorithms and data structures. • Apply important algorithmic design paradigms and methods of analysis. • Synthesize efficient algorithms in common engineering design situations.

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.

Module	Course Content	No. of Hours	POs mapped	PSOs mapped
Module-1	**Introduction:** Definition and characteristics of Algorithms; Analyzing algorithms; Program performance: time and space complexity, Asymptotic notation, complexity analysis. Recurrence equations and their solutions.	**08**	PO1/ PO2	PSO1/ PSO2
Module-2	**Sorting and order Statistics** Heap sort, Quick sort, Sorting in Linear time, Medians and Order Statistics. **Advanced Data Structure**: Red-Black Trees, Augmenting Data Structure. BTrees, Binomial Heaps, Fibonacci Heaps, Data Stricture for Disjoint Sets.	**08**	PO1/ PO2	PSO1/ PSO2
Module-3	**Dynamic Programming** with Examples Such as Knapsack. All Pair Shortest Paths – Warshal's and Floyd's Algorithms, Resource Allocation Problem. Backtracking, Branch and Bound with Examples Such as Travelling Salesman Problem, Graph Coloring, n-Queen Problem, Hamiltonian Cycles and Sum of Subsets.	**08**	PO1/ PO2	PSO1/ PSO2
Module-4	**Divide and Conquer** with Examples Such as Sorting, Matrix Multiplication, Convex Hull and Searching. Greedy Methods with Examples Such as Optimal Reliability Allocation, Knapsack, **Minimum Spanning Trees** – Prim's and Kruskal's Algorithms, Single Source Shortest Paths - Dijkstra's and Bellman Ford Algorithms.	**08**	PO1/ PO2/ PO3/PO 4	PSO1/ PSO2
Module-5	**Infeasibility:** P and NP classes; NP-hard problems Parallel algorithms: Introduction, data and control parallelism, parallel algorithms for matrix multiplication; embedding of problems graphs into processor graphs, load balancing and scheduling problems.	**08**	PO1/ PO2	PSO1/ PSO2
Total No. of Hours		**40**		

Learning Outcomes:	• Understand basic techniques for designing algorithms, including the techniques of recursion, divide-and-conquer, and greedy. • Design new algorithms, prove them correct, and analyze their asymptotic and

	absolute runtime and memory demands. • Apply classical sorting, searching, optimization and graph algorithms. • Explain the major graph algorithms and their analyses. Employ graphs to model engineering problems, when appropriate. Synthesize new graph algorithms and algorithms that employ graph computations as key components, and analyze them. • Analyze randomized algorithms. Employ indicator random variables and linearity of expectation to perform the analyses. Recite analyses of algorithms that employ this method of analysis.

Suggested books:

S. No.	Name of Authors /Books /Publisher/Year
1.	Sahni S, Data structures, Algorithms and applications in C++, McGraw Hill
2.	Aho, A.V., Hopcroft, J.E. & Ullman, J.D, The Design and Analysis of Computer algorithms, PHI
3.	Mchugh J.A., Algorithmic Graph Theory, PHI
4.	Quinn M.J., Parallel Computing Theory & Practice, McGraw Hill
5.	Goodman, S.E. &Hedetniemi, Introduction to the Design and Analysis of Algorithms, McGraw Hill

CO-PO/PSO MAPPING

Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	√	√											√	√
CO2	√	√	√										√	√
CO3	√	√	√										√	√
CO4	√	√	√	√								√	√	√
CO5	√	√	√									√	√	√